

# Mechanical Clock Wiring Guide

## 4-Module Digital Display with ESP32 Master Controller

### Project Overview

This guide covers the complete wiring for a modular mechanical clock system consisting of:

- **Master Module:** ESP32-S3 Super Mini with PCA9306 level shifter
- **4 Digit Modules:** Each containing PCF8575, ULN2003, 28BYJ-48 stepper, A3144 hall sensor, and status LEDs
- **Modular Bus System:** JST-XH connectors for easy assembly/disassembly

### System Architecture

#### Power Distribution

- **5V Rail:** Stepper motors, ULN2003 drivers, PCF8575s, Hall sensors, Module LEDs, PCA9306 VREF2
- **3.3V Rail:** ESP32 (internal regulator), PCA9306 VREF1, Master status LED

#### Communication

- **I2C Bus:** ESP32 ↔ PCA9306 ↔ All PCF8575 modules
- **Level Translation:** 3.3V (ESP32) ↔ 5V (modules) via PCA9306

### JST-XH Bus System

#### 4-Pin Connector Pinout

Pin 1: 5V (Red wire)  
Pin 2: GND (Black wire)  
Pin 3: SDA (Blue wire)  
Pin 4: SCL (Yellow wire)

#### Bus Topology (Daisy Chain)

[5V Supply] → [Master] → [Module 1] → [Module 2] → [Module 3] → [Module 4]

#### Cable Specifications

- **Power wires (Pins 1,2):** 22 AWG minimum
- **Signal wires (Pins 3,4):** 26 AWG
- **Length between modules:** 10-20cm each

- **Master to Module 1: 15-30cm**
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## Master Module Wiring

### Components Required

- ESP32-S3 Super Mini
- PCA9306 bidirectional level shifter
- 1x Blue/Green LED (status)
- JST-XH 4-pin connector
- Resistors: 2x 4.7k $\Omega$ , 2x 4.7k $\Omega$ , 1x 330 $\Omega$
- Capacitors: 2x 100nF ceramic

### Master Module Connections

#### Power Distribution

5V Supply Input → ESP32 VIN, PCA9306 VREF2, JST-XH Pin 1  
GND → All component grounds, JST-XH Pin 2

#### ESP32 Connections

ESP32 3.3V out → PCA9306 VREF1  
ESP32 GPIO21 (SDA) → PCA9306 SDA1  
ESP32 GPIO22 (SCL) → PCA9306 SCL1  
ESP32 GPIO2 → 330 $\Omega$  resistor → Status LED anode  
Status LED cathode → GND

#### PCA9306 Level Shifter

VREF1 → ESP32 3.3V output  
VREF2 → 5V supply  
SDA1 → ESP32 GPIO21  
SCL1 → ESP32 GPIO22  
SDA2 → JST-XH Pin 3  
SCL2 → JST-XH Pin 4  
GND → Common ground

#### Pull-up Resistors

4.7k $\Omega$ : ESP32 3.3V → PCA9306 SDA1  
4.7k $\Omega$ : ESP32 3.3V → PCA9306 SCL1  
4.7k $\Omega$ : 5V → PCA9306 SDA2  
4.7k $\Omega$ : 5V → PCA9306 SCL2

## Decoupling Capacitors

100nF: Across PCA9306 VREF1 to GND  
100nF: Across PCA9306 VREF2 to GND

## Digit Module Wiring

### Components Required (Per Module)

- PCF8575 GPIO expander
- ULN2003 stepper driver board
- 28BYJ-48 stepper motor
- A3144 hall effect sensor
- 1x Red LED (homing status)
- 1x Yellow/Orange LED (activity status)
- 2x JST-XH 4-pin connectors (input/output)
- Resistors: 1x 10k $\Omega$ , 2x 470 $\Omega$
- Capacitor: 1x 100nF ceramic

### Module Connections

#### JST-XH Bus Pass-Through

Input Connector Pin 1 (5V) → Output Connector Pin 1, PCF8575 VDD, ULN2003 VCC, Hall VCC  
Input Connector Pin 2 (GND) → Output Connector Pin 2, All component grounds  
Input Connector Pin 3 (SDA) → Output Connector Pin 3, PCF8575 SDA  
Input Connector Pin 4 (SCL) → Output Connector Pin 4, PCF8575 SCL

### PCF8575 GPIO Expander

VDD → 5V from JST connector  
VSS → GND  
SDA → JST connector Pin 3  
SCL → JST connector Pin 4

Pin Assignments:

P0 → ULN2003 IN1  
P1 → ULN2003 IN2  
P2 → ULN2003 IN3  
P3 → ULN2003 IN4  
P4 ← Hall sensor OUT (with 10k $\Omega$  pull-up to 5V)  
P5 → 470 $\Omega$  → Red LED anode (homing status)  
P6 → 470 $\Omega$  → Yellow LED anode (activity status)  
P7 → Available for future use

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## ULN2003 Stepper Driver

VCC → 5V  
GND → Ground  
IN1-IN4 → PCF8575 P0-P3  
OUT1-OUT4 → 28BYJ-48 stepper motor

## A3144 Hall Effect Sensor

VCC → 5V  
GND → Ground  
OUT → PCF8575 P4 (with 10kΩ pull-up to 5V)

## Status LEDs

Red LED (Homing):  
PCF8575 P5 → 470Ω resistor → LED anode  
LED cathode → GND  
  
Yellow LED (Activity):  
PCF8575 P6 → 470Ω resistor → LED anode  
LED cathode → GND

## Decoupling Capacitor

100nF: Across PCF8575 VDD to VSS

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## I2C Address Configuration

### PCF8575 Address Settings

Each module requires a unique I2C address set via A0, A1, A2 pins:

Module 1 (Hours Tens): A2=Low, A1=Low, A0=Low → Address 0x20  
Module 2 (Hours Units): A2=Low, A1=Low, A0=High → Address 0x21  
Module 3 (Minutes Tens): A2=Low, A1=High, A0=Low → Address 0x22  
Module 4 (Minutes Units): A2=Low, A1=High, A0=High → Address 0x23

### Address Pin Connections

- **Low:** Connect to GND
  - **High:** Connect to 5V (or leave floating - internal pull-up)
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## Power Budget & Current Analysis

### 5V Rail Current (Per Module)

- 28BYJ-48 stepper motor: ~200mA
- PCF8575: ~2mA
- A3144 hall sensor: ~10mA
- 2x Status LEDs: ~20mA
- **Per module total: ~232mA**

### Total System Current

- 4 modules:  $4 \times 232\text{mA} = 928\text{mA}$
- PCA9306: ~1mA
- ESP32 (5V input): ~100mA
- **5V rail total: ~1030mA**
- 3.3V rail (from ESP32): ~21mA

### Power Supply Requirements

- **5V supply:** Minimum 1.5A capacity recommended
  - **3.3V:** Supplied by ESP32 internal regulator (adequate)
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## Status LED Behavior

### Master Status LED (Blue/Green)

- **Solid ON:** System running normally
- **Fast blink (2Hz):** Initializing/connecting to modules
- **Slow blink (0.5Hz):** Error state (I2C communication issues)
- **OFF:** System not powered/crashed

### Module Homing LED (Red)

- **ON:** Homing sequence active
- **OFF:** Normal operation/homing complete

### Module Activity LED (Yellow/Orange)

- **Quick flash:** Stepper movement command received
  - **Solid ON:** Stepper currently moving
  - **OFF:** Stepper idle
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## Assembly Checklist

## Before Starting

- ☐ All JST-XH connectors and cables prepared
- ☐ Color-coded wires: Red(5V), Black(GND), Blue(SDA), Yellow(SCL)
- ☐ All components and resistor values verified
- ☐ Multimeter ready for continuity testing

## Master Module Assembly

- ☐ ESP32-S3 Super Mini wired
- ☐ PCA9306 level shifter installed
- ☐ Pull-up resistors (4x 4.7k $\Omega$ ) installed
- ☐ Status LED and current limiting resistor (330 $\Omega$ ) installed
- ☐ Decoupling capacitors (2x 100nF) installed
- ☐ JST-XH output connector wired
- ☐ Power connections tested (5V, GND)
- ☐ I2C signal continuity verified

## Per Module Assembly

- ☐ PCF8575 I2C address set correctly (A0, A1, A2)
- ☐ JST-XH input and output connectors wired
- ☐ PCF8575 to ULN2003 connections (P0-P3 to IN1-IN4)
- ☐ Hall sensor wired with pull-up resistor (10k $\Omega$ )
- ☐ Status LEDs with current limiting resistors (2x 470 $\Omega$ )
- ☐ Decoupling capacitor (100nF) installed
- ☐ Stepper motor connected to ULN2003 outputs

## System Integration

- ☐ Modules connected in daisy chain order
  - ☐ Power supply connected to master module
  - ☐ All connections tight and secure
  - ☐ No short circuits (continuity test)
  - ☐ I2C bus pull-ups present on master module only
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## Troubleshooting Guide

### No Communication with Modules

1. Check 5V and GND on JST-XH bus
2. Verify PCA9306 power (3.3V and 5V sides)
3. Confirm I2C pull-up resistors on master module
4. Test individual modules by connecting directly to master

## Module Not Responding

1. Verify unique I2C address settings
2. Check PCF8575 power and ground
3. Disconnect other modules to isolate issue
4. Verify JST-XH connector pinout and wiring

## Stepper Motor Issues

1. Check ULN2003 power (5V)
2. Verify PCF8575 P0-P3 to ULN2003 IN1-IN4 connections
3. Confirm motor wiring to ULN2003 OUT1-OUT4
4. Test motor independently with known good driver

## Hall Sensor Problems

1. Verify 5V power to sensor
2. Check 10kΩ pull-up resistor to 5V
3. Confirm sensor output to PCF8575 P4
4. Test sensor with magnet and multimeter

## Status LED Issues

1. Check current limiting resistor values
  2. Verify LED polarity (anode to resistor, cathode to GND)
  3. Test PCF8575 output pins with multimeter
  4. Replace suspected faulty LEDs
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## Component Specifications Summary

### ICs and Modules

- **ESP32-S3 Super Mini:** 3.3V logic, built-in 5V→3.3V regulator
- **PCA9306:** Bidirectional I2C level shifter (3.3V ↔ 5V)
- **PCF8575:** 16-bit I2C GPIO expander, 5V operation
- **ULN2003:** Darlington driver array, 5V/500mA per channel
- **28BYJ-48:** 5V stepper motor, ~200mA operation
- **A3144:** Hall effect sensor, 3.3V-5V operation

### Passive Components

- **Pull-up resistors:** 4.7kΩ (I2C bus, hall sensor)
- **LED resistors:** 330Ω (3.3V), 470Ω (5V)

- **Decoupling capacitors:** 100nF ceramic
  - **JST-XH connectors:** 4-pin, 3A rating
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## Final Notes

### Design Benefits

- **Modular:** Each digit module is self-contained and removable
- **Scalable:** Easy to add more modules or modify configuration
- **Serviceable:** Individual modules can be disconnected for repair
- **Clean:** Standardized JST-XH bus eliminates wire mess
- **Robust:** Proper level shifting and power distribution

### Safety Considerations

- Use appropriate wire gauges for current capacity
- Ensure all connections are secure before applying power
- Test continuity before first power-on
- Monitor current consumption during initial testing
- Use strain relief on all cable connections

### Software Considerations

- Each PCF8575 module requires unique I2C address
  - Implement proper I2C error handling and timeouts
  - Use status LEDs to provide visual feedback during operation
  - Consider implementing module auto-discovery for robustness
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